

References

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Papers introducing the method

- Moka, S., Liquet, B., Zhu, H. and Muller, S. (2024). *COMBSS: best subset selection via continuous optimization*. **Statistics and Computing** 34, 75. [doi:10.1007/s11222-024-10387-8](https://doi.org/10.1007/s11222-024-10387-8)
- Mathur, A., Liquet, B., Muller, S. and Moka, S. (2026). *Parsimonious Subset Selection for Generalized Linear Models with Biomedical Applications*. **arXiv preprint** [arXiv:2603.21952](https://arxiv.org/abs/2603.21952).

Software

- **R** — `combss` on CRAN — [package page](#)
- **Python** — `combss` on PyPI — [package page](#)
- Source for this presentation — github.com/saratmoka/combss-statfest

Datasets used in this presentation

- **Khan SRBCT**. Khan, J. et al. (2001). *Classification and diagnostic prediction of cancers using gene expression profiling and artificial neural networks*. **Nature Medicine** 7, 673–679. [link](#)
- **Rice GWAS**. McCouch, S. R. et al. (2016). *Open access resources for genome-wide association mapping in rice*. **Nature Communications** 7, 10532. [link](#)

Related methods cited

- **Lasso.** Tibshirani, R. (1996). *Regression shrinkage and selection via the Lasso*. **Journal of the Royal Statistical Society, Series B** 58 (1), 267–288.
- **MIO best subset.** Bertsimas, D., King, A. and Mazumder, R. (2016). *Best subset selection via a modern optimization lens*. **Annals of Statistics** 44 (2), 813–852.
- **SCAD.** Fan, J. and Li, R. (2001). *Variable selection via nonconcave penalized likelihood and its oracle properties*. **Journal of the American Statistical Association** 96 (456), 1348–1360.
- **MCP.** Zhang, C.-H. (2010). *Nearly unbiased variable selection under minimax concave penalty*. **Annals of Statistics** 38 (2), 894–942.

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